
Algorithm 1 One trial of localized rejection sampling with a step-level verifier in our study. The reported accuracy is the mean of the returned indicator over T independent trials.

input : step accuracy p ; solution step H ; verifier TPR t and TNR n ; false accept rate f ; round budget R

output: 1 if the final solution is fully correct, else 0

$x \leftarrow (x_1, \dots, x_H)$ with each $x_j \sim \text{Bernoulli}(p)$ // $x_j=1$: step j is correct

for $r \leftarrow 1$ **to** R **do**

 draw $u \sim \text{Uniform}(0, 1)$ **if** $x_j = 1$ **for all** j **then**

 // solution is fully correct

if $u < t$ **then return** 1 // correct accept

$k \leftarrow$ uniform random step in $\{1, \dots, H\}$ // false alarm

else

$e \leftarrow \min\{j : x_j = 0\}$ // first incorrect step

if $u < n$ **then**

$k \leftarrow e$ // correct localization

else if $u < n + (1 - n)f$ **then**

return 0 // false-accept

else

$k \leftarrow$ uniform random step in $\{1, \dots, H\} \setminus \{e\}$ // mislocalization

end

end

 resample $x_j \sim \text{Bernoulli}(p)$ for $j = k, \dots, H$ // keep prefix $x_{1:k-1}$, redo the suffix

end

return $\mathbf{1}[x_j = 1 \text{ for all } j]$
